MIL-A-43229A

10 June 1968
SUPERSEDING
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MILITARY SPECIFICATION

* ADHESIVE APPLYING MACHINES, LABEL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope.- This specification covers hand-operated labeling machines and hand-operated and electric motor driven adhesive applying machines for applying adhesive to paper labels.

* 1.2 Classification.- The machines covered by this specification shall be of the following types and styles (see 6.2):

Type I - Labeling Machine, Hand-Operated
Type II - Adhesive Applying Machine, Label (Gluing)

Style A - Hand-Operated
Style B - Motor Driven

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

FEDERAL

CC-M-636	-	Motors, Fractional Horsepower (Alternating Current).
QQ-B-626	-	Brass, Leaded and Non-Leaded: Rods, Shapes, Forging and Flat Products With Finished Edges (Bars, Flat Wire, and Strip).
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Conta Grade), and Cut Shapes.

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MIL-P-116	-	Preservation, Methods Of.

FSC 3540

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of.

(Copies of specifications, standards and publication required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

Standard Specification A167-63 - Corrosion-Resisting, Chromium-Nickel Steel Plate, Sheet, and Strip

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

UNDERWRITERS' LABORATORIES, INC.

Standard UL 73-1967 - Motor Operated Appliances

(Copies of Underwriters' Laboratories, Inc. publications may be obtained from the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 161 Sixth Street, New York, New York 10003 or 1655 Scott Boulevard, Santa Clara, California 95050.)

3. REQUIREMENTS

* 3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample of the adhesive applying machines, label or labeling machine covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

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* 3.2 Standard product.- The adhesive applying machines covered by this specification shall be the manufacturer's standard product except for any changes necessary to comply with specification requirements.

* 3.3 Compliance.- Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the electric motor driven adhesive applying machines for applying adhesive to paper labels he proposes to furnish under this specification meet applicable requirements of UL Standard No. 73-1967. Acceptable evidence of meeting applicable requirements of UL Standard No. 73-1967 shall be the UL label on the machine, a copy of the publishing UL Listing Card with re-examination service, a listing in the UL Electrical Appliance and Utilization Equipment List, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the machine conforms to applicable requirements of UL Standard No. 73-196

* 3.4 Materials.- Where materials are not definitely specified, they shall be of the quality normally used for that purpose in commercial practice.

3.4.1 Brass.- Brass rods shall conform to QQ-B-626.

* 3.4.2 Corrosion-resisting steel.- Corrosion-resisting steel shall conform to the chemical composition of any type of steel in the 300 series as specified in ASTM A167.

3.5 Design.-

* 3.5.1 Type I.- The type I labeling machine shall consist essentially of a gluing unit, frame, tank, and a handle. The type I machine shall be hand operated and designed to apply adhesive, cut, and attach a preprinted label, having a maximum width dimension of 3 inches, to any flat surface in one operation.

* 3.5.2 Type II.- The type II adhesive applying machine, label shall be hand operated or motor driven consisting essentially of a gluing unit, feed table, frame, scraper, scraper support, frame, and tank or reservoir. The type II machine shall be capable of applying adhesive to any size label having dimensi from 1/2 inch by 2 inches minimum to 8-1/2 inches width by any length. The width of the roller shall not exceed 9 inches. A dial regulator or screw adju ment shall regulate the adhesive layer which is applied to the label.

3.6 Construction.-

* 3.6.1 Type I.- The type I machine shall be so constructed that it may be used by either a right or left handed operator.

3.6.2 Type II.-

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* 3.6.2.1 Type II, style A.- The type II, style A machine feed roller and the gluing roller shall be so positioned that the feed roller shall not become soiled when in operation. The clearance between the bearings and the sides of the rollers shall be of sufficient size so as not to allow the mixing of the glue with oil from the bearing. Provisions shall be made to permit the removal of the tank or reservoir or gluing unit for cleaning.

* 3.6.2.2 Type II, style B.- The type II, style B machine shall meet the requirements of 3.6.2.1 except that the machine shall be motor driven and shall be provided with an on-off switch. The motor of the machine shall be as specified in 3.6.3.1. The drive between the motor and the gluing unit on a belt driven machine shall be fully inclosed.

3.6.3 Electrical requirement.-

* 3.6.3.1 Motor.- The motor on the type II, style B machine shall have a minimum of 1/20 horsepower rating for belt driven machines, or a minimum of 1/50 horsepower rating for direct coupled machines and shall conform to CC-M-636.

* 3.6.3.2 Electrical rating.- Unless otherwise specified (see 6.2), the type II, style B machine shall operate on a nominal 120 volt, single phase, 60 cycle, alternating current power system.

* 3.6.4 Feed table.- The feed table of the type II machine shall be fabricated from aluminum or corrosion-resisting steel specified in 3.4.2 and the table top shall have a commercial finish in accordance with the manufacturer's standard practice.

3.6.5 Frame.- The frame of the type I machine shall be fabricated from aluminum. The frame of the type II machine shall be fabricated from either bronze, aluminum or carbon steel.

* 3.6.6 Tanks or reservoirs.- The tank of the type I machine shall be fabricated from aluminum and shall have a capacity in accordance with the manufacturer's standard commercial product. The tank or reservoir of the type II machine shall be fabricated from bronze, aluminum, carbon steel, or cast iron and shall have a minimum capacity of 24 fluid ounces. Tanks or reservoirs fabricated from cast iron or carbon steel shall be painted inside and outside with an epoxy resin paint.

3.6.7 Bearings.- The bearings of the type II machine shall be fabricated from either bronze or oilite. The machine shall be constructed to allow for replacement of the bearings.

* 3.6.8 Feed roller.- The feed roller of the type II machine shall be fabricated from a bronze rod or brass tubing with a corrosion-resisting steel shaft.

* 3.6.9 Gears.- When gearing is required, the gears of the type I machine shall be aluminum, cycloidal, high speed type.

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* 3.7 Performance.- The type I and II machines shall be capable of applying adhesive to a label without oozing at the edges of the label and the label shall adhere securely without curling the edges when tested as specified in 4.4.1. The applied label shall be clean.

* 3.8 Electromagnetic compatibility.- Equipment procured for Army shall be designed and equipped for electromagnetic compatibility in accordance with Class IIB of MIL-STD-461 (see 4.4.2). When specified (see 6.3), equipment procured for Navy and Air Force shall be in accordance with Class IIB of MIL-STD-461 (see 4.4.2).

* 3.9 Marking for identification.- Each type and style of machine shall be marked in accordance with MIL-STD-130.

* 3.10 Finish.- Each type and style of machine shall be finished in accord ance with the manufacturer's standard commercial practice.

* 3.11 Workmanship.- The type I and II machine shall be clean and well mad All sharp burrs, slivers, or splinters shall be removed. All bolts, nuts, screws, and studs shall be drawn up tight. Wiring shall be properly joined shall not be cut or abraded; wiring shall not be loose at terminal ends or otherwise impaired.

4. QUALITY ASSURANCE PROVISIONS

4.1 Repsonsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any othe facilities suitable for the performance of the inspection requirements speci fied herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, the preproduction sample shall be examined in accordance with the defects in table I and specified dimensions, and shall be tested in accordance with 4.4.1 and 4.4.2 when applicable.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordanc with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with requirements of referenced subsidiary specifications and standards to the extent specified o if none, in accordance with this specification.

* 4.3.2 End item inspection.- All adhesive applying machines, label and labeling machines, of the same type and style, offered for inspection at one time shall be considered a lot. The sample unit for these inspections shall one completely fabricated machine.

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* 4.3.3 Visual examination.- The completely assembled end item shall be examined for defects as classified in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not finished as specified.		X
Design	Characteristic not in accordance with the specified requirements.	X	
Construction and workmanship (general)	Part missing or not specified type.	X	
	Fractured, punctured, dented, deteriorated, bowed, malformed, or otherwise impaired.	X	
	Misplaced or not in proper alignment.	X	
	Part loose.		X
	Sharp burr, sliver, or splinter.	X	
	Functioning component that requires abnormal force to operate.		X
	Not connected as specified.	X	
	Adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended.	X	
	Component not accessible for servicing where required.	X	
	Glue oozing beyond edges of label.	X	
	Applied label not clean.	X	
Bolts, nuts, screws or other types of threaded fasteners	Missing, broken, stripped, or loose.		X
	Component not properly assembled or secured.	X	
	Unit out of square or alignment.	X	
Identification marking and equipment manual	Missing, incomplete, not legible.		X

* 4.3.4 End item testing.- When a preproduction sample is not required, the initial production unit shall be tested as specified in 4.4.1 and 4.4.2 when applicable.

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* 4.3.5 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, packing, and marking, as required by section 5 of this specification, are complied with. Defects will be scored as specified in table II. The lot size shall be the number of containers in the inspection lot. The sample unit shall be one shipping contain fully prepared for delivery. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Material	Component missing, damaged, or otherwise defective.
Workmanship	Bulging or distortion of containers.
Weight	Exceeds specified weight.

* 4.3.6 When containers are required to be in accordance with PPP-B-636 examination for closure and reinforcing shall be in accordance with the appendix of PPP-B-636.

4.4* Tests.-

4.4.1 Performance test.-

* 4.4.1.1 Type I.- The type I machine shall be capable of applying the correct amount of adhesive, cut, and attach 10 labels to a newspaper or magazine to determine compliance with 3.7.

* 4.4.1.2 Type II.- The type II machine shall be capable of applying the correct amount of adhesive to ten labels and the operator shall place the labels on a carton material to determine compliance with 3.7.

4.4.2 Electromagnetic compatibility test.- When electromagnetic compatibili is required, the preproduction sample or initial unit of production shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or his private testing organization. The supplier shall furnish the contracting officer for approval the Interference Control Plan, the EMI/EMC Test Plan and the Electromagnetic Compatibility Test Report required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of the specification are met, the preproduction sample, or initial unit of production as applicable, shall be used as a model for all production units.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A.- Each machine shall be packaged in a fiberboard box conforming to style RSC, type CF or SF, class domestic of PPP-B-636. Prior to packaging motors, openings into control and junction boxes shall be sealed with tape conforming to class 3 of PPP-T-60. The machine shall be immobilized within the box with die-cut or scored fiberboard inserts. The equipment manual shall be packaged Method IC-1 of MIL-P-116 and placed in the fiberboard box with the machine. Each box shall be securely closed in accordance with the appendix of the box specification.

* 5.1.2 Level C.- Each machine shall be packed to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

* 5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A.- Machines, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with strapping or tape banding in accordance with the appendix of the container specification. The weight of contents of each shipping container shall not exceed 65 pounds.

* 5.2.2 Level B.- Machines, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to Style RSC, type CF or SF, class domestic Grade 275 of PPP-B-636. Each box shall be closed in accordance with Method II of the Appendix of PPP-B-636. The weight of the contents of each shipping container shall not exceed 65 pounds.

* 5.2.2.1 When specified (see 6.2), the shipping container shall conform to V3c or V3s of PPP-B-636 and closure shall be in accordance with the class weather-resistance requirement of the appendix of PPP-B-636. The shipping container material may also be V4s of PPP-F-320.

5.2.3 Level C.- Machines, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

* 6.1 Intended use.- The type I machine covered by this specification is intended for applying adhesive, cutting, and attaching label to any flat surface in one operation. The type II machine covered by this specificatio is intended for applying adhesive to labels.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and style machine required (see 1.2).
(c) First article.- When a preproduction sample is not required (see 3.1).
(d) Electrical requirements, if other than specified (see 3.6.3).
(e) When electromagnetic compatibility is required for Navy and Air Force use (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(g) Whether weather-resistant fiberboard containers are required for level B shipments (see 5.2.2.1).

* 6.3 Electromagnetic compatibility.- The contracting officer should submi copy of the Interference Control Plan, the EMI/EMC Test Plan, and the test r port required (see 4.4.2), to the Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey 07703 for approval.

* 6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Review activities:

Army - SM, EL, MD
Navy - SH, SA
Air Force - 85

User activities:

Army - SM, ME

Preparing activity:

Army - GL

Project No. 3540-0050

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